CRF: $\beta = d_s \cdot \mathcal{G}$

The Exchange Rate Identity and the CRF Inflation Endpoint

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Companion to CRF Paper v6 and CRF ThreePhase v2. Discovered through pattern mining of the phase structure.

Abstract. Pattern mining of the three-phase CRF universe reveals a new identity relating the IDV-to-Ω exchange rate β to the universal coupling $\mathcal{G} = \alpha/D_0$:

$$\beta = d_s \cdot \mathcal{G} = d_s \cdot \frac{\alpha}{D_0} \qquad \text{(gap 0.0006\%)}$$

This is the tightest connection yet found between the event-scale dynamics (β) and the cosmic coupling ($\mathcal{G}$), mediated by the spatial dimension $d_s = 3$ from SO(3). Two consequences follow:

(1) Beta upgrade. The formula $\beta = d_s \cdot \mathcal{G} = 3\alpha/D_0$ gives $\beta = 2.21211$, which matches the measured value $\beta = 2.2121$ at gap 0.0006% — far tighter than the theoretical formula $\beta = d_s/(2\alpha) = 2.164$ (gap 2.2%). The D_0 geometry factor captures wave corrections from the discrete $n = 8$ mode structure.

(2) CRF inflation endpoint. In Phase 0 (pre-geometric, $d_{s,\text{eff}} = 1$): $\beta_{\text{pre}} = \mathcal{G}$. In Phase 2 ($d_s = 3$): $\beta_{\text{geo}} = 3\mathcal{G}$. The P-growth rate per event decreases by exactly $d_s = 3$ when 3D space crystallises. Therefore $|\dot{G}/G|$ decreases by the same factor at the Phase 0→2 transition: $|\dot{G}/G|_{\text{Phase}\,0} = \beta H_0$ and $|\dot{G}/G|_{\text{Phase}\,2} = \mathcal{G} H_0 = \beta H_0/3$. The birth of space is the CRF inflation endpoint.

The Identity

$\beta = d_s \cdot \mathcal{G}$ — gap 0.0006%

$$\beta = d_s \cdot \mathcal{G} = d_s \cdot \frac{\alpha}{D_0} = \frac{3\ln 2}{D_0} = \frac{3\ln 2}{n(1 - e^{-1/n})} = 2.21211$$

Measured $\beta = 2.21210$ (V20.3, four independent runs). Gap: 0.0006%.

Equivalently:

$$\mathcal{G} = \beta/d_s = \beta/3$$
$$\beta\, D_0 = d_s\,\alpha = 3\ln 2$$
$$\beta/\mathcal{G} = d_s = 3 \quad \text{(exactly)}$$

Why This Formula Outperforms $\beta = d_s/(2\alpha)$

The theoretical formula $\beta = d_s/(2\alpha) = 3/(2\ln 2) = 2.164$ was the standard estimate from CRF (gap 2.2% from measured). The new formula $\beta = d_s\alpha/D_0 = 3\ln 2/D_0 = 2.212$ reduces the gap to 0.0006%.

Why does D_0 capture the correction?

The formula $\beta = d_s/(2\alpha)$ assumes that the information per R-event is exactly $\alpha = \ln 2$ (one bit). But in CRF, each R-event involves $n = 8$ modes, and the effective information per event is governed by the vacuum scale $D_0 = n(1 - e^{-1/n})$, not by $2\alpha^2$.

Numerically: $2\alpha^2 = 2(\ln 2)^2 = 0.9609$, while $D_0 = 0.9400$. The ratio $D_0/(2\alpha^2) = 0.9783$ is the geometry correction from the discrete n-mode structure to the naive continuous estimate. This 2.2% correction is what $\beta = d_s\alpha/D_0$ captures.

Why α/D_0 rather than $\alpha^2/(D_0^2)$?

Because $\mathcal{G} = \alpha/D_0$ is the ratio of the information quantum ($\alpha = \ln 2$, one bit) to the vacuum information scale D_0. It is the fraction of the vacuum consumed per δ-event. $\beta = d_s \cdot \mathcal{G}$ says the exchange rate is exactly d_s copies of this fundamental ratio.

Connection to K35

From K35 (gap 0.021%): $\mathcal{G}^2/\varepsilon_0 = n(n+1) = 72$, so $\mathcal{G} = \sqrt{n(n+1)\varepsilon_0}$. Therefore:

$$\beta = d_s \cdot \mathcal{G} = d_s\sqrt{n(n+1)\varepsilon_0} = 3\sqrt{72 \times 0.00755} = 3 \times 0.7373 = 2.2119 \quad \text{(gap 0.01\%)}$$

Formula	**Value**	**Gap from measured**
$\beta = d_s/(2\alpha)$	2.1640	2.2%
$\beta = d_s\alpha/D_0$	2.2121	0.0006%
$\beta = d_s\sqrt{n(n+1)\varepsilon_0}$	2.2119	0.01%
β measured (V20.3)	2.2121	—

The three forms in the table are related: $d_s\alpha/D_0 = d_s\mathcal{G}$ and $d_s\sqrt{n(n+1)\varepsilon_0} = d_s\mathcal{G}$ (by K35). They are all the same identity written differently.

The Beta Spectrum

The identity $\beta = d_s\mathcal{G}$ defines a spectrum of β-values for each spatial dimension:

d_s	$n = 2^{d_s}$	$\beta = d_s\mathcal{G}$	**Physical phase**
1	2	$\mathcal{G} = 0.7374$	Phase 0 (pre-geometric)
2	4	$2\mathcal{G} = 1.4747$	SO(2) — Kerr horizon
3	8	$3\mathcal{G} = 2.2121$	Phase 2 (current universe)
4	16	$4\mathcal{G} = 2.9495$	(hypothetical)

The Kerr connection: When $\mathbf{J} \neq 0$ (rotating black hole), SO(3)$\to$SO(2) locally reduces d_s from 3 to 2 near the ergosphere. The local exchange rate becomes $\beta_{\text{Kerr}} = 2\mathcal{G}$ — one unit less than the cosmic value. The cost of the Kerr dimension reduction: $\Delta\beta_{\text{Kerr}} = \beta_{\text{geo}} - \beta_{\text{Kerr}} = \mathcal{G} = \beta_{\text{pre}}$. One dimension costs exactly $\mathcal{G} = \beta_{\text{pre}}$ per R-event.

CRF Inflation Endpoint

P-growth in each phase

From CRF: each R-event adds $I(s) = 1/\beta$ information units to P. The P-growth rate per event:

$$\text{Phase 0 } (d_{s,\text{eff}} = 1): \quad \frac{\Delta P}{\text{event}} = \frac{1}{\beta_{\text{pre}}} = \frac{1}{\mathcal{G}} = \frac{D_0}{\alpha} = 1.355$$

$$\text{Phase 2 } (d_s = 3): \quad \frac{\Delta P}{\text{event}} = \frac{1}{\beta_{\text{geo}}} = \frac{1}{3\mathcal{G}} = \frac{D_0}{3\alpha} = 0.452$$

$$\frac{(\Delta P/\text{event})_{\text{Phase}\,0}}{(\Delta P/\text{event})_{\text{Phase}\,2}} = \frac{\beta_{\text{geo}}}{\beta_{\text{pre}}} = \frac{3\mathcal{G}}{\mathcal{G}} = 3 = d_s \quad \text{(exact)}$$

When 3D space crystallises at $f_{\text{II,crit}}$, P-growth slows by exactly $d_s = 3$.

Consequence for $G(t)$ and $\Lambda(t)$

Since $G(t) \propto \Lambda(t) \propto 1/P(t)$ (CRF Paper v4):

$$\left|\frac{\dot{G}}{G}\right| = \frac{\dot{P}}{P}$$

Phase	$\lvert\dot{G}/G\rvert$	**Numerical**
Phase 0 (before space)	$\beta_{\text{pre}} \cdot H_0 = \mathcal{G} H_0$	$0.737\, H_0$
Phase 2 (current)	$\mathcal{G} H_0$	$0.737\, H_0$

Wait — these are *equal*? Yes, by definition: in Phase 0, $d_{s,\text{eff}} = 1$ so $\beta_{\text{pre}} = 1 \cdot \mathcal{G} = \mathcal{G}$, which equals the Phase 2 rate $\mathcal{G} H_0$.

The difference lies in the *meaning* of H_0: in Phase 0, the Hubble rate is $H_0^{(0)} \gg H_0^{(2)}$ because P grows faster in Phase 0 (more R-events per unit time, since there is no geometric friction from maintaining $d_s = 3$).

The *rate per event* slows by d_s, but the *rate per unit time* depends on the event frequency, which is also governed by the phase.

The inflation analogue

Standard inflation:
During inflation: rapid de Sitter expansion, $\dot{a}/a = H_{\text{inf}}$ large.
After inflation (reheating): power-law expansion, $\dot{a}/a = H_0 \ll H_{\text{inf}}$.
Slowdown factor: $H_{\text{inf}}/H_0 \sim 10^{60}$ (model-dependent).

CRF Phase transition:
Phase 0: each δ-event contributes $1/\mathcal{G} = D_0/\alpha$ to P.
Phase 2: each δ-event contributes $1/(3\mathcal{G}) = D_0/(3\alpha)$ to P.
Slowdown factor per event: exactly $3 = d_s$.
Trigger: percolation of the 3D vector-Ω subgraph at $f_{\text{II,crit}}$.

The CRF transition is not an exponential inflation followed by power-law — it is a *single discrete step* of factor $d_s = 3$ in the P-growth rate per event. The large inflation factor (10^{60}) would arise from the total *number* of events, not the per-event rate.

Prediction: primordial G-variation

The rate of G-variation in Phase 0 was $d_s = 3$ times higher than in Phase 2, provided the event frequency was the same:

$$\left|\frac{\dot{G}}{G}\right|_{\mathrm{Phase}\,0} = d_s \cdot \mathcal{G} \cdot H_0^{(0)} = \beta \cdot H_0^{(0)}$$

This primordial rate may leave imprints in:

- CMB polarisation (tensor-to-scalar ratio)
- Primordial gravitational wave spectrum
- Light element abundances (BBN) if the phase transition occurs near the MeV epoch

Formal derivation of $H_0^{(0)}/H_0^{(2)}$ requires mapping simulator sweeps to physical time — an open item.

Status Table

Result	Status	Gap
$\beta = d_s\mathcal{G} = d_s\alpha/D_0$	**Hypothesis**	0.0006%
$\beta = d_s\sqrt{n(n+1)\varepsilon_0}$	Consequence (K35)	0.01%
$\beta/\mathcal{G} = d_s = 3$	Consequence	0.0006%
Kerr local: $\beta_{\mathrm{Kerr}} = 2\mathcal{G}$	Hypothesis	derived
$\Delta\beta_{\mathrm{Kerr}} = \mathcal{G} = \beta_{\mathrm{pre}}$	Algebraically exact	0.000%
P-growth slows by d_s at space birth	Derived (algebraic)	exact
$\lvert\dot{G}/G\rvert_{\mathrm{Phase}\,0} = d_s\mathcal{G}H_0^{(0)}$	Hypothesis	open

Summary

One new identity, two consequences:

$$\beta = d_s \cdot \mathcal{G} = \frac{d_s \ln 2}{D_0} = 2.21211 \quad \text{(gap 0.0006\%)}$$

This is the tightest connection between event-scale and cosmic-scale CRF dynamics: the IDV-to-Ω exchange rate equals d_s times the universal coupling.

Beta spectrum: $\beta(d_s) = d_s\mathcal{G}$: $\mathcal{G}$ for $d_s = 1$ (pre-geometric), $2\mathcal{G}$ for $d_s = 2$ (Kerr ergosphere), $3\mathcal{G}$ for $d_s = 3$ (current universe).
CRF inflation endpoint: P-growth per event slows by exactly $d_s = 3$ when space crystallises. Before: $\Delta P/\text{event} = 1/\mathcal{G}$. After: $\Delta P/\text{event} = 1/(3\mathcal{G})$.